

## Project Development Phase

### Coding & Solution

|               |                                 |
|---------------|---------------------------------|
| Date          | 10/07/2026                      |
| Name          | Jaswanth Madapuram              |
| Project Name  | Credit Card Approval Prediction |
| Maximum Marks | 5 Marks                         |

### Functional Features Overview:

This document lists all the functional features implemented in the **Credit Card Approval Prediction** project. Each feature contributes to building an intelligent system that predicts whether a credit card application should be approved based on applicant information using machine learning techniques.

| S.No | Feature Name                    | Feature Description                                                          | Module / Component   | Status (Done / In Progress / Pending) | Marks Contribution |
|------|---------------------------------|------------------------------------------------------------------------------|----------------------|---------------------------------------|--------------------|
| 1    | Data Collection                 | Import and load the credit card application dataset.                         | Data Module          | Done                                  | High               |
| 2    | Data Preprocessing              | Handle missing values, encode categorical features, and normalize data.      | Preprocessing Module | Done                                  | High               |
| 3    | Feature Engineering             | Select relevant features and prepare data for model training.                | Feature Engineering  | Done                                  | Medium             |
| 4    | Model Training                  | Train machine learning models such as Logistic Regression and Random Forest. | ML Model             | Done                                  | High               |
| 5    | Credit Card Approval Prediction | Predict whether a customer's application will be approved or rejected.       | Prediction Module    | Done                                  | High               |

|   |                  |                                                                                |                       |      |        |
|---|------------------|--------------------------------------------------------------------------------|-----------------------|------|--------|
| 6 | Model Evaluation | Evaluate performance using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. | Evaluation Module     | Done | Medium |
| 7 | User Interface   | Allow users to input applicant details and receive prediction results.         | Flask Web Application | Done | High   |
| 8 | Result Display   | Display prediction outcome with confidence and user-friendly messages.         | Output Module         | Done | Medium |

### Feature Summary:

| Metric                             | Count / Value |
|------------------------------------|---------------|
| Total Features Planned             | 8             |
| Total Features Implemented         | 8             |
| Core / Must-Have Features          | 6             |
| Additional / Nice-to-Have Features | 2             |
| Features Tested & Verified         | 8             |

### Feature Category Breakdown:

| S.No | Category                  | Features in Category | Example Features                                                           |
|------|---------------------------|----------------------|----------------------------------------------------------------------------|
| 1    | User Interface (UI)       | 2                    | Applicant details form, Prediction result page                             |
| 2    | Backend / Logic           | 4                    | Data preprocessing, Feature engineering, Model training, Prediction engine |
| 3    | Database / Storage        | 1                    | Dataset loading, Saved ML model (Joblib/Pickle)                            |
| 4    | API / Integration         | 1                    | Flask backend connecting UI with prediction model                          |
| 5    | Security / Authentication | 0                    | Not applicable for the current project                                     |